The Rope Winding Angle as the Weinberg Angle

A Geometric Estimate $\sin^2\theta_W = 1/(3\sqrt{2})$, and Its ~2% Gap

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Abstract

The rope framework proposes that the electroweak mixing (Weinberg) angle is set by the winding geometry of the two-strand rope, giving $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$ — the '3' from the three spatial dimensions and the '√2' from the two strands. This estimate sits about 1.94% from the measured value 0.23122. We report it as a geometric conjecture of the correct magnitude and structure, and we are explicit that the ~2% gap is real and consequential: it is the input that, when used in the lepton-mass relation, breaks it. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants.

Scope of this paper

CLAIM: a geometric estimate $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$ from two-strand rope winding, ~2% from measurement.

NOT CLAIMED: a precision derivation. The ~1.94% gap is real; the value is a conjecture and the sector's weak link. Status: Modeled (Conjecture).

1. Winding Geometry and the Mixing Angle

The Weinberg angle parametrises the mixing of the electroweak gauge fields. In the rope framework it is read from the winding geometry of the two-strand rope: three spatial dimensions supply the factor 3, the two strands supply the √2, giving $\sin^2\theta_W = 1/(3\sqrt{2})$.

$$\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357 \quad (1.1)$$ — measured: 0.23122

2. The Gap, Stated Plainly

The honest status of this value. The predicted $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$ differs from the measured 0.23122 by about 1.94%. It is close, and its form is suggestive, but it is NOT exact, and the test suite pins it explicitly as a 'soft external input' — the value that, when fed into the lepton-mass relation instead of the measured angle, breaks it. This is therefore a geometric conjecture of the right magnitude, not a precision derivation, and the ~2% gap is the known weak link of the electroweak sector.

Status

$\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$ geometric estimate — Modeled (Conjecture).
~1.94% deviation from measured 0.23122 — the known weak link; breaks lepton masses if used.

Registered Claims in This Paper
Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

• **EW-001 [Conjecture]:** Weinberg angle sin2thetaW = 1/

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.